

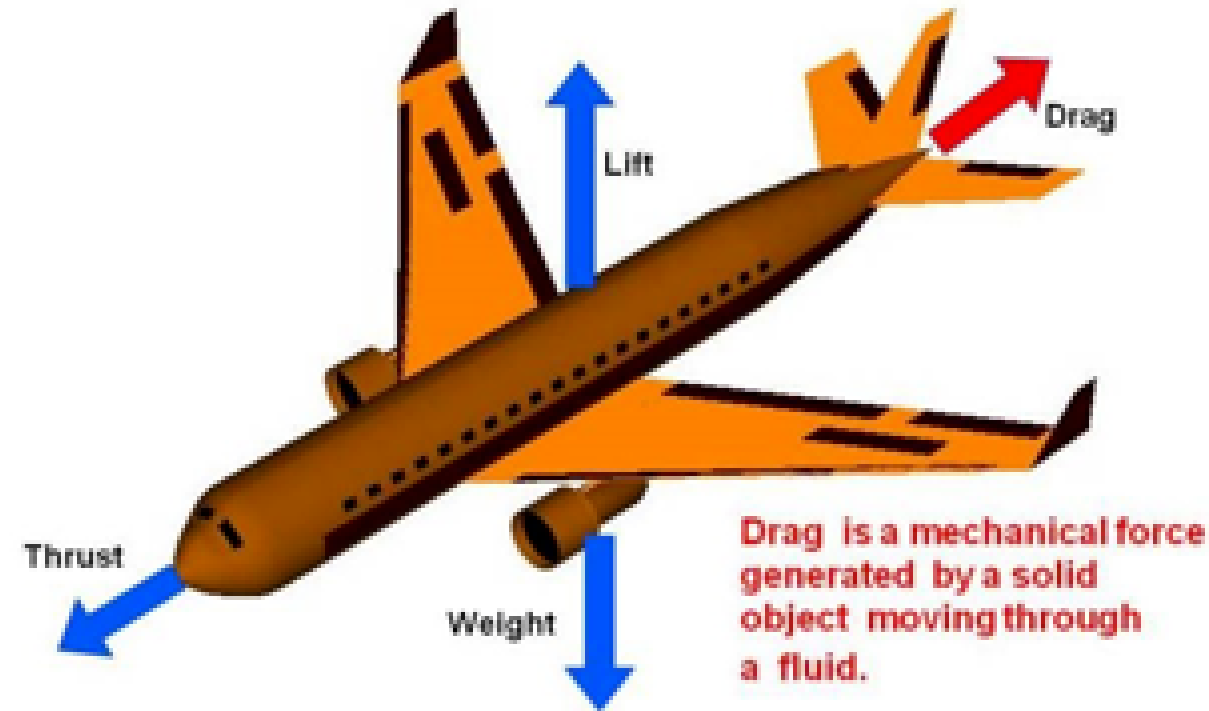
Day 07 - Drag Forces



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Announcements

- Homework 2 is due Friday (late Sunday)
- Homework 3 is now posted
- Office hours, for now (Mihir-MN; Danny-DC):
 - Thursday 3-5pm (MN, 1248 BPS)
 - Friday 10-12pm (DC, 1248 BPS)
 - Friday 3-5pm (MN, Zoom only)
- Zoom Link for class: <https://msu.zoom.us/j/99550311023>
 - password: phy321msu

Seminars this week

WEDNESDAY, January 28, 2026

Astronomy Seminar, 1:30pm, BPS 1400 & Zoom

Speaker: Rebecca Kyer & Emily Elizondo, MSU

Title: TBA

Zoom Link: [https://msu.zoom.us/j/93334479606?](https://msu.zoom.us/j/93334479606?pwd=OtIXPWhRPBfzYu53sl3trSJlaBYI7C.1)

[pwd=OtIXPWhRPBfzYu53sl3trSJlaBYI7C.1](https://msu.zoom.us/j/93334479606?pwd=OtIXPWhRPBfzYu53sl3trSJlaBYI7C.1)

Passcode: 825824

Seminars this week

WEDNESDAY, January 28, 2026

Physics Education Research Seminar, 3:00pm, BPS 1400 & Zoom

Speaker: Jennifer Doherty, MSU

Title: Principle-based reasoning: A strategy for developing expertise in Physiology

Zoom Link: <https://msu.zoom.us/j/96470703707>

Passcode: PERSeminar

Seminars this week

WEDNESDAY, January 28, 2026

FRIB Nuclear Science Seminar, 3:30pm, Zoom Only

Speaker: Gwen Grinyer, University of Regina

Title: Precision Spectroscopy of Rare Isotopes

Zoom Link: [https://msu.zoom.us/j/91051885898?](https://msu.zoom.us/j/91051885898?pwd=S423bks5tzaeOwNb1tLaUaHDUScm5A.1)

[pwd=S423bks5tzaeOwNb1tLaUaHDUScm5A.1](https://msu.zoom.us/j/91051885898?pwd=S423bks5tzaeOwNb1tLaUaHDUScm5A.1)

Passcode: 949110

Seminars this week

FRIDAY, January 30, 2026

QuIC, Seminar, 12:40 p.m., BPS 1300 & Zoom

Speaker: Ben DalFavero, MSU

Title: Fault tolerant quantum computing II

*For the full schedule, please see: <https://sites.google.com/msu.edu/quic-seminar/> or for more information, please reach out to Ryan LaRosa directly

Seminars this week

FRIDAY, January 30, 2026

IReNA Online Seminar, 2:00 pm, Zoom

Light refreshments at 1:50pm in 2025 Nuclear Conference Room - FRIB

Hosted by: Aldana Grichener (University of Arizona & Observatory)

Speaker: Mengke Li, University of California, Berkeley

Title: Implications of a Weakening $N = 126$ Shell Closure Away from Stability
for r-Process Astrophysical Conditions

Zoom Link: <https://msu.zoom.us/j/827950260>

Password: CENAM

Goals for Week 3

- Be able to answer the following questions.
 - What is Mathematical Modeling?
 - What is the process for analyzing these models?
- Be able to solve "Simple" Motion Problems with Newton's Laws.

Our man, Reynolds

The Reynolds number is a dimensionless quantity. It is a ratio of inertial forces to viscous forces.

$$Re = \frac{\rho v L}{\mu}$$

- ρ - density of the fluid
- v - velocity of the object
- L - characteristic length
- μ is the dynamic viscosity



Our man, Reynolds

BTW, this is not a photo of Reynolds. **This is Stokes.**

- He developed the concept of the Reynolds number.
- Reynolds "popularized" it according to the Wikipedia.

$$Re = \frac{\rho v L}{\mu}$$

Discussion: What kinds of systems have a high/low Reynolds number?



Clicker Question 7-1

Consider a ball falling down with a coordinate system such that y is positive downward. Which of the following is correct for the differential equation?

1. $m\ddot{y} = -mg + bv_y$

2. $m\ddot{y} = -mg - bv_y$

3. $m\ddot{y} = +mg + bv_y$

4. $m\ddot{y} = +mg - bv_y$

Think about the SIGN of v_y ; it's a component not a speed!

Assuming a **linear model** for Air Resistance $\sim bv$, we can obtain this EOM for the velocity a falling ball, after *separation of variables*:

$$\frac{dv_y}{v_y - v_{term}} = -\frac{b}{m} dt$$

Let's try to integrate this equation from $v_y = v_{y0}$ at $t = 0$ to $v_y(t)$ at time t .

We can write the integral (on both sides) that is needed to solve this equation.

This integral setup is common in physics derivations with ODEs; provided we can find an anti-derivative.

Clicker Question 7-2

Assuming a **quadratic model** for Air Resistance $\sim cv^2$, what is the EOM for a falling ball, with y positive upward?

1. $m\ddot{y} = -mg + cv_y^2$

2. $m\ddot{y} = -mg - cv_y^2$

3. $m\ddot{y} = +mg + cv_y^2$

4. $m\ddot{y} = +mg - cv_y^2$

5. ???

Draw a picture, remember v_y is a component, and consider upward and downward motion!

Clicker Question 7-3

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = (mg/c)^{1/2}$. Do the units make sense? What are the units $[gt/v_{term}]$?

1. Yes, **they have the same units**; the units for $[gt/v_{term}]$ are m/s.
2. No, **they have different units**; the units for $[gt/v_{term}]$ are m/s.
3. Yes, **they have the same units**; the units for $[gt/v_{term}]$ are unit-less.
4. No, **they have different units**; the units for $[gt/v_{term}]$ are unit-less.

Clicker Question 7-4

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = \sqrt{mg/c}$. What happens when $t \rightarrow \infty$?

1. The object stops moving.
2. The object travels at a constant velocity.
3. The object travels at an increasing velocity.
4. The object travels at a decreasing velocity.
5. I'm not sure.

Clicker Question 7-5

Assuming a **quadratic model** for Air Resistance $\sim cv^2$, what is the EOM for a ball initially sent upward and to the right, with y positive upward and x positive to the right?

1. $m\ddot{x} = -cv_x\sqrt{v_x^2 + v_y^2}; \quad m\ddot{y} = -mg - cv_y\sqrt{v_x^2 + v_y^2}$

2. $m\ddot{x} = -cv_x\sqrt{v_x^2 + v_y^2}; \quad m\ddot{y} = +mg - cv_y\sqrt{v_x^2 + v_y^2}$

3. $m\ddot{x} = +cv_x\sqrt{v_x^2 + v_y^2}; \quad m\ddot{y} = -mg + cv_y\sqrt{v_x^2 + v_y^2}$

4. $m\ddot{x} = +cv_x\sqrt{v_x^2 + v_y^2}; \quad m\ddot{y} = +mg + cv_y\sqrt{v_x^2 + v_y^2}$

5. None of the above.

Clicker Question 7-6

For linear drag in 2D, we found the following equations of motion:

$$\dot{v}_x = -\gamma v_x$$

$$\dot{v}_y = -g - \gamma v_y$$

Are these equations integrable? *Can we find anti-derivatives for both equations?*

1. Yes, both equations are integrable.
2. No, neither equation is integrable.
3. Only the x equation is integrable.
4. Only the y equation is integrable.